

# Hackathon Problem Statements

Regd No: 25091A32C6, 25091A32G5

Problem Statement No: 103

**Gig-Economy Worker Payment Reconciliation System** Build a payment reconciliation system for gig-economy platforms using an OOP-based transaction ledger that tracks worker earnings across multiple jobs, platform fees, and payout schedules transparently. Flag discrepancies using automated reconciliation logic. This addresses growing concerns about payment transparency and fair compensation for India's expanding gig workforce.

Regd No: 25091A32F9, 25091A32H8

Problem Statement No: 113

**Building Structural Safety Scorer** Create a structural safety assessment tool using a rule-based OOP inspection checklist system that scores buildings against safety parameters (age, material, maintenance history, code compliance) to flag high-risk structures needing urgent inspection. This supports municipal building safety departments in prioritizing limited inspection resources toward the most dangerous structures first.

Regd No: 25091A32K2, 25091A32P6

Problem Statement No: 129

**Domestic Violence Helpline Call Pattern Analyzer** Build an aggregate-level, fully anonymized call pattern analyzer for domestic violence helplines using clustering to identify temporal and regional trends (without exposing any individual case data), helping policymakers allocate support resources like shelters and counselors more effectively to high-need areas. Use OOP to structure aggregated statistical modules only, with strict privacy-by-design principles.

Regd No: 25091A3266, 25091A3245

Problem Statement No: 132

Corruption Risk-Flagging System for Public Tenders Create an anomaly detection tool that flags unusual patterns in public tender/bid datasets (single-bidder tenders, price clustering near estimates, repeat winner patterns) using statistical outlier detection to support transparency audits. Use OOP to model tenders, bidders, and risk flags. This supports anti-corruption oversight bodies in prioritizing tenders for deeper investigation.

Regd No: 25091A3289, 25091A3293

Problem Statement No: 135

Government Job Vacancy vs Applicant Skill Matcher Build a matching system using classification and clustering to compare government job vacancy requirements against a pool of applicant skill profiles, identifying best-fit candidates and highlighting systemic skill gaps in the applicant pool. Use OOP to model jobs, applicants, and matching scores. This can make public sector recruitment more efficient and evidence-based.

Regd No: 25091A32H4, 25091A32C9

Problem Statement No: 138

Water Tanker Distribution Scheduler for Drought-Hit Areas Build a water tanker scheduling system using greedy and priority algorithms to allocate limited water tanker trips to the most water-stressed localities first, based on reported shortage severity and population data. Use OOP to model tankers, localities, and distribution schedules. This ensures fairer water distribution during acute drought or supply-disruption periods.